DSA-0308- NATURAL LANGUAGE PROCESSING

ASSIGNMENT 3

CO3: CONTEXT-FREE GRAMMARS AND PARSING

COURSE: NATURAL LANGUAGE PROCESSING

NATURE: INDIVIDUAL ASSIGNMENT

TOTAL MARKS: 50

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REGISTER NUMBER: 192424240

CONTENTS AND SUBMISSION NOTE

This report answers the tasks in the corresponding uploaded assignment. It follows the task order and preserves the assignment's terminology. The worked examples explain the reasoning and the evidence expected.

Important: where the assignment requires a corpus experiment, annotation, code execution or human judgement, no unrun result is presented as an actual measurement. Such values must be collected by running the experiment and inserted into the report. Any sample calculation is explanatory only.

Task 1 Context-Free Grammar

Concepts and direct answer

Define G=(N, Sigma, R, S). Use non-terminals S, NP, VP, PP, AdjP, Det, N, V, P and Conj; terminals are helpdesk words. Example rules: S->NP VP; NP->Det N; NP->NP PP; VP->V NP; PP->P NP; S->S Conj S. Include at least 40 rules and a lexicon of at least 60 words. Explain constituency, heads and phrase categories using substitution, movement and coordination tests. CFG rules rewrite one non-terminal independently of context.

The answer begins by identifying the input, the target interpretation and the assumptions. Terms are defined before they are used. This prevents the same label from being applied to different meanings in different parts of the analysis.

The main conclusion is that the system must make its intermediate decisions explicit. A final label or output alone is not enough to show that the method is correct.

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Task 1 Context-Free Grammar

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Task 1 Context-Free Grammar

Worked example / trace

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Task 1 Context-Free Grammar

Evaluation, limitations and conclusion

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Task 2 Derivations and Parse Trees

Concepts and direct answer

A derivation expands non-terminals until only terminals remain. Leftmost derivation always expands the leftmost non-terminal; rightmost derivation expands the rightmost. Show both for five sentences, bracketed notation and tree diagrams. State tree height, node count and rule applications with a consistent counting convention. Demonstrate PP-attachment ambiguity with two trees. A sentence can have multiple derivation orders that correspond to the same tree.

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Task 3 Sentence Constructions

Concepts and direct answer

Represent declaratives with S->NP VP, imperatives with an understood subject, yes/no questions with auxiliary inversion, and wh-questions with a fronted wh-phrase and a gap. Add subordinate and relative clauses, coordination and long-distance dependencies. Show one parse tree for each construction. Explain how each added construction increases grammar size and possible ambiguity.

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Task 4 Agreement

Concepts and direct answer

Agreement includes subject-verb person and number, determiner-noun number and pronoun case. A multiplied grammar uses categories such as NP-sg, VP-sg, NP-pl and VP-pl, causing rule growth as features combine. Feature-based rules unify NUM and PER values and reject conflicts. Test at least six grammatical and six ungrammatical sentences and tabulate decisions.

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Task 5 Subcategorization

Concepts and direct answer

Subcategorization is the complement pattern selected by a verb. Include intransitive crash, transitive submit, ditransitive send, sentential complement tell and PP complement apply for. Encode frames with distinct verb categories, then discuss category multiplication. Distinguish complements from optional adjuncts. Show acceptance and rejection of under- and over-complemented examples.

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Task 6 Top-Down Parsing

Concepts and direct answer

A recursive-descent parser starts from S, expands non-terminals, matches terminals and backtracks on failure. Give pseudocode and traces for one unambiguous and one ambiguous sentence. Left recursion such as NP->NP PP can cause infinite recursion. Eliminate left recursion or use a left-corner filter. Discuss repeated subparses and worst-case growth with k PPs.

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Task 7 Earley Parsing

Concepts and direct answer

An Earley state is a dotted rule plus origin index. Predictor adds productions when the dot precedes a non-terminal; Scanner consumes a matching terminal; Completer advances states waiting for a completed constituent. Hand-simulate at least six words and list chart sets S[0] through S[n]. Explain backpointers and packed ambiguity. General CFG worst-case complexity is O(n^3) time and O(n^2) chart space.

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Task 8 Feature Structures

Concepts and direct answer

Represent lexical entries with features NUM, PER, CASE and SUBCAT. Unification merges compatible feature structures; conflicting values cause failure. Define subsumption and reentrancy. Augment grammar rules with feature constraints and show a successful complete parse plus one failed unification. Explain integration during Earley completion and added computational cost.

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Task 9 PCFG

Concepts and direct answer

A PCFG assigns P(A->beta|A), with alternatives for each left-hand side summing to one. Estimate rule probabilities by maximum likelihood from a mini-treebank: count(rule)/count(lhs). Calculate probabilities of two ambiguous parse trees and sum tree probabilities for sentence probability. Explain CNF conversion and probabilistic CKY/Viterbi. Discuss independence assumptions and lexical sensitivity.

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Task 9 PCFG

Worked example / trace

A PCFG assigns P(A->beta|A), with alternatives for each left-hand side summing to one. Estimate rule probabilities by maximum likelihood from a mini-treebank: count(rule)/count(lhs). Calculate probabilities of two ambiguous parse trees and sum tree probabilities for sentence probability. Explain CNF conversion and probabilistic CKY/Viterbi. Discuss independence assumptions and lexical sensitivity.

Worked trace template: (1) identify the input units; (2) list candidate interpretations or structures; (3) apply the declared rule; (4) record the selected result and any unresolved alternative; (5) verify the result against a hand-worked example or gold annotation.

When two interpretations remain plausible, retain both or ask a clarification question. A forced choice without evidence can create a downstream error.

Evidence	Required content
Input	Sentence, corpus, query or scenario
Method	Formula, grammar, rule or algorithm
Trace	Intermediate steps and selected output
Validation	Metric, counterexample or comparison

Task 9 PCFG

Evaluation, limitations and conclusion

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Evaluation should include a normal case, a boundary case and an adversarial case. Report expected behaviour, actual behaviour, error category and a specific refinement. Use the same test inputs when comparing two designs.

Conclusion: select a design only when its measured behaviour meets the task requirements. Discuss remaining limitations and the conditions under which a simpler alternative may be preferable.

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Task 10 Implement and Test

Concepts and direct answer

Implement grammar, top-down parser, Earley parser and PCFG/Viterbi parser. Test 30 sentences including ambiguity, left recursion, ungrammatical and long inputs. Record parse found, number of parses, chart states/steps and runtime. Compare with hand calculations and plot states against sentence length. Keep fixed data, grammar and seed.

The answer begins by identifying the input, the target interpretation and the assumptions. Terms are defined before they are used. This prevents the same label from being applied to different meanings in different parts of the analysis.

The main conclusion is that the system must make its intermediate decisions explicit. A final label or output alone is not enough to show that the method is correct.

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Representation and method

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A reliable design separates input preparation, representation, decision procedure and output. Each stage should have a stated data structure and a clear rule for updating it. Keep the original input so that normalisation does not erase information needed later.

For reproducibility, record the data source, assumptions, parameter values, software version and fixed random seed wherever applicable.

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Task 11 Compare and Recommend

Concepts and direct answer

Compare top-down, Earley and probabilistic parsers on coverage, ambiguity, cost and maintenance. A feature-based CFG with Earley parsing and PCFG ranking is a suitable controlled-helpdesk recommendation, with a partial parsing fallback for noisy input. Discuss manual grammar and annotation effort and remaining limitations.

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Task 12 Design Challenge

Concepts and direct answer

New messages contain embedded questions, coordination, relative clauses, ellipsis, fragments and missing punctuation. Trace at least three through the original parser and identify the first failure. Modify grammar and features, add robust partial parsing and update probabilities. Quantify coverage, ambiguity, chart size and annotation effort; defend retain/modify/redesign using evidence.

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Additional Analysis – Engineering trade-offs

Discuss accuracy, memory, latency, transparency, annotation cost and maintainability. A method is useful only relative to its operating constraints.

Application to this assignment: CO3: CONTEXT-FREE GRAMMARS AND PARSING. The design should preserve the input, expose intermediate decisions, state uncertainty and allow each stage to be checked independently.

Suggested evidence record: Input | Expected result | Actual result | Error category | Proposed correction. Complete this record using the student's own execution or annotation; do not present an illustrative value as a measured result.

CONCLUSION

The answers in this report follow the structure and concepts of the uploaded assignment. The central engineering lesson is to define representations and decision rules, trace the reasoning, test difficult cases, measure performance and defend the final design with evidence.

REFERENCES AND ACKNOWLEDGEMENT

Primary source: the corresponding DSA03 Natural Language Processing assignment PDF supplied for this task. Add exact textbooks, papers, corpora, toolkits and software references actually consulted, including author or organisation, year, title and publication details.

Disclose AI or reference tools used as required by the assignment. Verify the examples and replace all illustrative experimental values with results from your own implementation, corpus analysis or annotation.